

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
 CHAMOS Matrusanstha Department of Mechanical Engineering

ME – 416 OPERATION RESEARCH	
CO1	Students will be able to understand various quantitative techniques and their role in problem solving.
CO2	They will be able to select and apply tools and find appropriate solutions to decision making problems.
CO3	Students can recognize the importance and value of Operations Research and mathematical modeling in solving practical problems in industry.
CO4	Students can understand Operations Research models and apply them to real-life problems.
CO5	Students will be able to build and solve Transportation Models and Assignment Models.

List of Experiment (ME416 - OR)		
Sr. No.	Title	Course Outcomes
1	Introduction to Operation Research	CO3
2	Exercise on formulation of LP problems	CO3, CO4
3	Exercise on graphical solution of LP problems	CO1, CO4
4	Exercise on Simplex method	CO3, CO4
5	Exercise on Two-Phase method	CO3, CO4
6	Exercise on duality of LP problems	CO3, CO4
7	Exercise on Transportation problems	CO5
8	Exercise on Assignment problems	CO5
9	Exercise on Queuing problems	CO1, CO3, CO4
10	Exercise on Replacement problems	CO3, CO4
11	Exercise on Game Theory problems	CO2
12	Exercise on Dynamic Programming	CO1, CO3

List of Assignment (ME416 - OR)		
Sr. No.	Title	Course Outcomes
1	Assignment 1- Linear programming	CO3, CO4
2	Assignment 2- Transportation and Assignment	CO5
3	Assignment 3- Queuing and Replacement	CO1, CO3, CO4
4	Assignment 4- Games theory and Dynamic Programing	CO1, CO2, CO3

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Operation Research ME416

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Operation Research ME416

Experiment No 1 Introduction to Operation Research

AIM:

Introduction to Operation Research

THEORY:

Operation Research rather simply defined is the research of operations. An operation may be called a set of acts required for the achievement of a desired outcome. Such complex, inter-related acts can be performed by four types of systems: Man, Machine, Man-Machine unit and any organization of men, machines and man-machine units. OR is concerned with the operations of the last type of system.

OR in the most general sense, can be characterized as the application of scientific methods, tools and techniques to problems involving the operations of systems so as to provide those in control of the operations with optimum solutions to the problem.

QUESTIONS:

1. Give different definitions of operation research.
2. Give different characteristics of operation research.
3. Discuss the importance of operation research in decision making.
4. Discuss various phases of operation research in detail.
5. What are the advantages and limitation of operation research?
6. Give various applications of operation research.

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Experiment No 2

Exercise on formulation of LP problems

AIM:

To study how to formulate Linear Programming problems.

THEORY:

The linear programming method is the technique for choosing the best alternative from a set of feasible alternatives, in situations in which the objective function as well as the constraints can be expressed as linear mathematical functions.

QUESTIONS:

1. What are the requirements of linear programming problem?
2. What are the three components of a typical LP problem?
3. What are the applications of linear programming method?
4. The Agricultural Research Institute suggested to a farmer to spread out at least 4800 kg of a special phosphate fertilizer and not less than 7200 kg of a special fertilizer to raise productivity of crops in his fields. There are two sources for obtaining these-mixtures A and B. Both of these are available in bags weighing 100 kg each and they cost Rs. 40 and Rs. 24 respectively. Mixture A contains phosphate and nitrogen equivalent of 20 kg and 80 kg respectively, while mixture B contains these ingredients equivalent of 50 kg each. Write this as a linear programming problem.
5. A furniture manufacturer produces two types of desks: Standard and Executive. These desks are sold to an office furniture wholesaler, and for all practical purposes, there is an unlimited market for any mix of these desks, at least within the manufacturer's production capacity. Each desk has to go through four basic operations: cutting of the lumber, joining of the pieces, pre-finishing, and final finish. Each unit of the standard desk produced takes 48 minutes of cutting time, 2 hours of joining, 40 minutes of pre-finishing and 4 hours of final finishing time. Each unit of the executive desk required 72 minutes of cutting, 3 hours of joining, 2 hours of pre-finishing and 5 hours and 20 minutes of final finishing time. The daily capacity for each operation amounts to 16 hours of cutting, 30 hours of joining, 16 hours of pre-finishing and 64 hours of final finishing time. The profit per unit produced is Rs. 40 for the standard desk and Rs. 50 for the executive desk. Formulate the LP problem.

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Experiment No 3 Exercise on Graphical Solution of LP problems

AIM:

Study of graphical method of solution for LP problem.

THEORY:

A linear programming problem with only two variables presents a simple case, for which the solution can be derived using a graphical or geometrical method. Though in actual practice such small problems are rarely encountered, the graphical method provides a pictorial representation of the solution process and a great deal of insight into the basic concepts used in solving large LP problems.

QUESTIONS:

1. Solve the following LP problem graphically.

Maximize $W = 2y_1 + y_2$
subjected to $y_1 + 2y_2 \leq 10$
 $y_1 + y_2 \leq 6$
 $y_1 - y_2 \leq 2$
 $y_1 - 2y_2 \leq 1$
 $y_1, y_2 \geq 0$

2. Solve the following LP problem graphically.

Minimize $Z = 20x_1 + 10x_2$
subjected to $x_1 + 2x_2 \leq 40$
 $3x_1 + x_2 \geq 30$
 $4x_1 + 3x_2 \geq 60$
 $x_1, x_2 \geq 0$

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Experiment No 4 Exercise on Simplex method

AIM:

Study of Simplex method and Big-M method..

THEORY:

Simplex method, also called simplex technique or simplex algorithm was developed in 1947 by G. B. Dantzig, an American mathematician. It has the advantage of being universal i.e. any, linear model for which the solution exists, can be solved by it. In principle, it consists of starting with a certain solution of which all that we know is that it is basic feasible, i.e. it satisfies the constraints as well as non-negativity conditions ($x_j \geq 0$, $j = 1, 2, 3, \dots, n$). We, then, improve upon this solution at consecutive stages, until, after a certain finite number of stages, we arrive at the optimal solution. In Big-M method we assign coefficients to artificial variables, undesirable from the objective function point of view. If objective function Z is to be minimized then a very large positive price called penalty is assigned to each artificial variable. Similarly if z is to be maximized then a very large negative price also called penalty is assigned to each of these variables. The penalty will be designated by $-M$ for a maximization problem and $+M$ for a minimization problem where $M > 0$.

QUESTIONS:

1. Explain the following in short; (a) Basic feasible solution (b) Degenerate basic feasible solution (c) Optimal solution (d) Feasible solution (e) slack variable (f) Artificial variable (g) Surplus variable (h) unbounded solution (i) infeasible solution
2. What is degeneracy in simplex? Why it should be resolved?
3. A company makes two kinds of leather belts. Belt A is a high quality belt and belt B is of lower quality. The respective profits are Rs. 4 and Rs. 3 per belt. The production of each of type A requires twice as much time as a belt of type B, and if all belts were of type B, the company could make 1000 per day. The supply of leather is sufficient for only 800 belts per day (both A and B combined). Belt a requires a fancy buckle and only 400 per day are available. There are only 700 buckles a day available for belt B. What should be the daily production of each type of belt? Formulate this problem as an LP model and solve it by simplex method.
4. Solve the following by Big-M method.
Maximize $Z = 3x_1 - x_2$
subjected to $2x_1 + x_2 \geq 2$
 $x_1 + 3x_2 \leq 3$
 $x_2 \leq 4$
 $x_1, x_2 \geq 0$

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Experiment No 5 Exercise on Two-Phase method

AIM:

Study of Two-Phase method.

THEORY:

In the first phase of this method the sum of the artificial variables is minimized subject to the given constraints to get a basic feasible solution of the LP problem. The second phase minimizes the original objective function starting with the basic feasible solution obtained at the end of the first phase. Since the solution of the LP problem is completed in two phases, this is called the two-phase method.

QUESTIONS:

1. What are the advantages of two-phase method?
2. Solve the following by two-phase method
Maximize $Z = 5x + 3y$
subjected to $2x + y \leq 1$
 $x + 4y \geq 6$
 $x, y \geq 0$
3. Use two-phase simplex method to solve the following
Maximize $Z = 5x_1 + 8x_2$
subjected to $3x_1 + 2x_2 \geq 3$
 $x_1 + 4x_2 \geq 4$
 $x_1 + x_2 \leq 5$
 $x_1, x_2 \geq 0$

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Experiment No 6

Exercise on duality of LP problems

AIM:

Study of duality method in LP problems.

THEORY:

In the context of linear programming, duality implies that each linear programming problem can be analysed in two different ways but having equivalent solutions. Each LP problem both maximization and minimization stated in its original form has associated with another linear programming problem called dual linear programming problem or in short dual which is unique based on the same data. In general it is immaterial which of the two problems is called primal or dual, since dual of the dual is primal.

Metals/alloys used in engineering may be classified into categories: (i) Ferrous (ii) Non-ferrous. The ferrous metals/alloys are carbon based which include varieties of irons & steels.

QUESTIONS:

1. Enlist primal –dual relationship for all functions.
2. What is duality? Why is it preferable to solve the dual rather than the primal? Discuss with suitable example.
3. Write the dual of the following LP problem
Maximize $Z = 3x_1 + 5x_2 + 7x_3$
Subjected to $x_1 + x_2 + 3x_3 \leq 10$
 $4x_1 - x_2 + 2x_3 \geq 15$
 $x_1, x_2 \geq 0$
4. Solve the following LP problem by using its dual:
Maximize $Z = 5x - 2y + 3z$
subjected to $2x + 2y - z \geq 2$
 $3x - 4y \leq 3$
 $y + 3z \leq 5$
 $x, y, z \geq 0$

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Experiment No 7

Exercise on Transportation problems

AIM:

Study of Transportation problems.

THEORY:

Transportation models deals with problems concerning as to what happens to the effectiveness function when we associate each of a number of origins(sources) with each of a possibly different number of destinations. The total movement from each origin and the total movement to each destination is given and it is desired to find how the associations be made subject to the limitations on totals. In such problems, sources can be divided among the jobs or jobs may be done with a combination of sources. The distinct feature of transportation problems is that sources and jobs must be expressed in terms of only 1 kind of unit.

QUESTIONS:

1. List out the methods of obtaining IBFS. What conditions produce degeneracy in transportation problems? How it can be resolved?
2. Write the mathematical statement of transportation problem explaining each term involved in it.
3. What are dummy destinations (warehouse) and dummy origins (factories)? When are they used and why? How can you predict whether the solution is unique or alternate?
4. Solve the following problem:

To From	1	2	3	Supply
1	5	1	7	10
2	6	4	6	80
3	3	2	5	15
Demand	75	20	50	

Suppose there are penalty costs for every unsatisfied demand unit which are given by 5, 3 & 2 for destinations 1,2 & 3 respectively.

5. Based on the profit matrix, find most profitable schedule:

		Sales Agencies					Supply
		P ₁	P ₂	P ₃	P ₄	P ₅	
Factories	F ₁	15	17	12	11	11	140
	F ₂	5	9	7	15	7	190
	F ₃	14	15	16	20	10	115
Demand		74	94	69	39	119	

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Experiment No 8 Exercise on Assignment problems

AIM:

Study of Assignment problems.

THEORY:

An assignment problem concerns as to what happens to the effectiveness function when we associate each of a number of 'origins' with each of the same number of 'destinations'. Each resource or facility is to be associated with one and only one job and associations are to be made in such a way so as to maximize (or minimize) the total effectiveness. Resources are not divisible among jobs, nor are jobs divisible among resources. Given n facilities and n jobs and given the effectiveness of each facility for each job, the problem is to assign each facility to one and only one job so as to optimize the given measure of effectiveness.

QUESTIONS:

1. Distinguish between Transportation model and Assignment model.
2. Write the mathematical statement of an assignment & Traveling salesman problem
3. Give the applications of assignment problems.
4. Consider the problem of assigning five operators to five machines. The assignment costs are given below:

		Operators				
Machines		A	B	C	D	E
	M ₁	10	5	13	15	16
	M ₂	3	9	18	13	6
	M ₃	10	7	2	2	2
	M ₄	7	11	9	7	12
	M ₅	7	9	10	4	12

5. Solve the assignment problem presented by the following matrix for minimization:

	I	II	III	IV	V	VI
A	9	22	58	11	19	27
B	43	78	72	50	63	48
C	41	28	91	37	45	33
D	74	42	27	49	39	32
E	36	11	57	22	25	18
F	13	56	53	31	17	28

Also find out the another optimum solution if exist.

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Experiment No 9

Exercise on Queuing problems

AIM:

Study of Queuing methods.

THEORY:

A queue is formed when either units requiring service—commonly referred as customers, wait for service or the service facilities, stand idle and wait for customers. Some customers wait when the total number of customers requiring service exceeds the number of service facilities, some service facilities stand idle when the total number of service facilities exceeds the number of customers requiring service.

Queuing theory can be applied to a wide variety of operational situations where imperfect matching between the customers and service facilities is caused by one's ability to predict accurately the arrival and service times of customers. In particular, it can be used to determine the level of service (either the service rate or the number of service facilities) that balances the following two conflicting costs: (a) cost of offering the service (b) cost incurred due to delay in offering service.

QUESTIONS:

1. Explain the following terms related to queuing theory: (a) service time (b) idle time (c) rate of service (d) rate of interval (e) utilization factor (g) balking (h)reneging (i) jockeying
2. Discuss the field of applications for queuing. Explain queue discipline and its various forms.
3. Explain Kendal's notation for queuing model.
4. What do you understand by queuing structure? Explain (a) First-come First Served (b) Last-come First Served and (c) Service in random-order basis of customer handling.
5. What do you mean by traffic intensity?
6. Workers come to tool room to receive special tools. The average time between two intervals is 60 seconds & the arrivals are assumed to be in Poisson distribution. The average service time is 40 seconds. Determine
 - (a) average queue length
 - (b) Probability that the attendant remains idle
 - (c) Average number of workers in the system
 - (d) Mean waiting time of an arrival

7. A tailor is specialist in ladies dresses. The number of customer approaches the tailor appears to be Poisson distribution with a mean of 6 customers per hour. The tailor attends the customers on FCFS basis and customers wait if the need be. The tailor can attend the customers at an average rate of 10 customers per hour with service time exponentially distributed.

Find

- (a) Find the probability of the number of arrivals (0 through 5) during (i) a 15-minute interval, and (ii) a 30-minute interval
- (b) The utilization parameter
- (c) The probability that the queuing system is idle
- (d) Expected number of customers at tailor shop
- (e) How much time should a customer expect to spend in queue?
- (f) What is the expected time would a customer spend in a tailor shop?
- (g) The average time that the tailor is free on a 10-hour working day
- (h) The probability associated with the number of customers (0 through 5) in the queuing system
- (i) What is the expected number of customers waiting for tailor's services?
- (j) What is the average length of queues that have at least one customer?
- (k) Assuming that $n > 0$ (i.e. customers are in system) what is the probability that the waiting time (excluding service time) of a customer in the queue shall be more than 10 minutes?
- (l) Assuming that the customers are in the system, what is the probability that a customer shall be in the shop for more than 15 minutes?

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Experiment No 10

Exercise on Replacement problems

AIM:

Study of Replacement problems.

THEORY:

In any establishment, sooner or later all equipment need to be replaced. A replacement is called for whenever new equipment offers more efficient or economical service than the old, existing one. For example the old equipment might fail and work no more or is worn out and needs higher expenditure on its maintenance. The problem, in such situations, is to determine the best policy to be adopted with respect to replacement of the equipment. The replacement theory provides answer to this question in terms of optimal replacement period.

QUESTIONS:

1. What is replacement? Why does the problem of replacement arise? Briefly explain the relevant costs in replacement of depreciable assets.
2. Explain individual replacement and group replacement with suitable example.
3. Define (a) money value (b) present worth factor (c) discount rate.
4. The data on the operating costs per year and resale prices of the equipment A whose purchase price is Rs. 10,000 are given below:

Year	1	2	3	4	5	6	7
Operating costs (in Rs.)	1500	1900	2300	2900	3600	4500	5500
Resale value (in Rs.)	5000	2500	1250	600	400	400	400

- (a) What is the optimum period for replacement?
 - (b) When the equipment A is 2 years old the equipment B which is a new model for the same usage is available. The optimum period for replacement is 4 years with average costs of Rs. 3600. Should we change the equipment A with that of B? If so when?
5. The following mortality rates have been observed for a certain type of light bulbs:

Week	:	1	2	3	4	5
% failing by week end	:	10	25	50	80	100

There are 1000 bulbs & its cost Rs. 2 to replace an individual bulb which has burnt out. If all bulbs were replaced simultaneously it would cost 50 paise per bulb. It is proposed to replace all the bulbs at fixed intervals, whether or not they have burnt out & to continue replacing burnt out bulbs as they fail. At what intervals should all bulbs be replaced?

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Experiment No 11 Exercise on Game Theory

AIM:

Study of Game Theory.

THEORY:

Game theory is a body of knowledge which is concerned with the study of decision making in situations where two or more rational opponents are involved under conditions of competition and conflicting interests. It deals with human processes in which an individual decision making unit who can be an individual, a group, a formal or informal organization, or a society, is not in complete control of the other decision making units, the opponents and is addressed to problems involving conflict, cooperation or both at various levels. The main objective in the theory of games is to determine the rules of rational behavior in the game situations, in which the outcomes are dependent on the actions of the interdependent players. A game refers to a situation in which two or more players are competing.

QUESTIONS:

1. Explain “Max-Min.” and “Mini-Max” principle used in game theory with suitable illustration.
2. Define the following terms related to Game theory (a) pure strategy (b) saddle point (c) payoff matrix (d) zero sum game (e) mixed strategy.
3. Solve the following by rule of Dominance and find the value of game:

Player A		I	II	III	IV
	I	3	2	4	0
	II	3	4	2	4
	III	4	2	4	0
	IV	0	4	0	8

4. For the following pay-off matrix by using rule of Dominance, determine the best strategy and the value of game:

	Y		
	J	K	L
P	60	50	40
Q	70	70	50
R	80	60	75

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Experiment No 12

Exercise on Dynamic Programming

AIM:

Study of Dynamic Programming problems.

THEORY:

Often decision making process involves several decisions to be taken at different times. For example problems of inventory control, evaluation of investment opportunities, long-term corporate planning, and so require sequential decision-making. The mathematical technique of optimizing such a sequence of interrelated decisions over a period of time is called dynamic programming. It uses the idea of recursion to solve a complex problem, broken into a series of interrelated (sequential) decision stages (also called sub-problems) where the outcome of a decision at one stage affects the decision at each of the next stages. The word dynamic has been used because time is explicitly taken into consideration.

QUESTIONS:

1. Define dynamic programming.
2. Discuss the characteristic and applications of dynamic programming and the principle of optimality.
3. Discuss following terms of dynamic programming (a) stage (b) state (c) return function.
4. Use dynamic programming to solve the following LP problem

$$\text{Maximize } Z = 3x_1 + 5x_2$$

subjected to $x_1 \leq 4$

$$x_2 \leq 6$$

$$3x_1 + 2x_2 \leq 18$$

$$x_1, x_2 \geq 0$$

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